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APPENDIX*

SECOND QUANTIZATION

Let us consider a collection of N particles, contained in a box of volume Ω, on which we impose periodic boundary conditions. To describe the state of such a system, we ordinarily use the wave function $\psi(\mathbf{r}_1 \cdots \mathbf{r}_N)$, depending on the $3N$ coordinates of the particles in "configuration" space. Such a representation allows a natural generalization of methods applied to a single particle. It is not the only possible representation; there are others, much more practical.

Let us consider a complete set of *single-particle states*—for example, plane waves of momentum $\mathbf{p}$ (we ignore spin for the moment). One can specify the state of a system by indicating the number $n_\mathbf{p}$ of particles found in the "box" $\mathbf{p}$. We are thus led to choose as a basis for the space of *states of the total system* the vectors

$$|n_1, \ldots, n_\mathbf{p}, \ldots\rangle,$$

where each of the $n_\mathbf{p}$ can take on any positive integral value. (Note that the total number of particles is no longer conserved.) Such a state is simply a product (eventually symmetrized or antisymmetrized) of plane waves, one for each particle.

We can define a "creation" operator $a_\mathbf{p}^+$ by the relation

$$a_\mathbf{p}^+|\cdots, n_\mathbf{p}, \cdots\rangle = \sqrt{n_\mathbf{p}+1}\,|\cdots, n_\mathbf{p}+1, \cdots\rangle. \quad \text{(A.1)}$$

In the present representation, known as the "occupation-number" representation, $a_\mathbf{p}^+$ has a single nonzero matrix element. Its complex

* This appendix follows closely Appendix B of P. Nozières, *Theory of Interacting Fermi Systems*, W. A. Benjamin, New York (1963).

conjugate, or "destruction" operator, is defined by

$$a_{\mathbf{p}}|\cdots, n_{\mathbf{p}}, \cdots\rangle = \sqrt{n_{\mathbf{p}}}\,|\cdots, n_{\mathbf{p}}-1, \cdots\rangle. \tag{A.2}$$

Our objective is to express all the properties of the system in terms of these creation and destruction operators.

Let us first take a boson gas, and let us consider the product

$$a_{\mathbf{p}}^{+}a_{\mathbf{p}'}^{+}|\,\rangle.$$

We have created a particle $\mathbf{p}'$, then another, $\mathbf{p}$. The same result would be obtained by inverting the order, as the total wave function is invariant with respect to permutation of two particles. Hence we have the relation

$$[a_{\mathbf{p}}^{+}, a_{\mathbf{p}'}^{+}] = 0. \tag{A.3}$$

By the same method it is easily shown that

$$[a_{\mathbf{p}}, a_{\mathbf{p}'}] = 0, \qquad [a_{\mathbf{p}}, a_{\mathbf{p}'}^{+}] = \delta_{\mathbf{p}\mathbf{p}'}. \tag{A.4}$$

The operator

$$N_{\mathbf{p}} = a_{\mathbf{p}}^{+}a_{\mathbf{p}} \tag{A.5}$$

measures the number of particles in the state $\mathbf{p}$ (since $N_{\mathbf{p}}|\cdots, n_{\mathbf{p}}, \cdots\rangle = n_{\mathbf{p}}|\cdots, n_{\mathbf{p}}, \cdots\rangle$).

We now consider the fermion gas; we denote the creation operator for a particle of momentum $\mathbf{p}$ by $c_{\mathbf{p}}^{+}$, the destruction operator by $c_{\mathbf{p}}$, and proceed to build up a set of states in a fashion directly analogous to the boson case. The difference between the two systems is manifested in the symmetry properties of the many-body wave function. That for fermions must be antisymmetric on the interchange of any two particles; the operators $c_{\mathbf{p}}$ must therefore satisfy *anticommutation relations*,

$$[c_{\mathbf{p}}^{+}, c_{\mathbf{p}'}^{+}]_{+} = [c_{\mathbf{p}}, c_{\mathbf{p}'}]_{+} = 0, \qquad [c_{\mathbf{p}}^{+}, c_{\mathbf{p}'}]_{+} = \delta_{\mathbf{p}\mathbf{p}'}. \tag{A.6}$$

It can be seen from Eq. (A.6) that $c_{\mathbf{p}}^{2} = 0$; a state can accommodate only a single particle. Moreover, the occupation number operator

$$N_{\mathbf{p}} = c_{\mathbf{p}}^{+}c_{\mathbf{p}}$$

is such that $N_{\mathbf{p}}^{2} = N_{\mathbf{p}}$ [as can be verified with the help of Eqs. (A.6)]. Therefore, $N_{\mathbf{p}}$ can equal only 0 or 1.

We note that for fermions, the equations analogous to (A.1) and (A.2) read

$$c_{\mathbf{p}}|\cdots, n_{\mathbf{p}}, \cdots\rangle = (-1)^{\sum_{j<p} n_j}\, n_{\mathbf{p}}|\cdots, n_{\mathbf{p}}-1, \cdots\rangle, \tag{A.1a}$$

$$c_{\mathbf{p}}^{+}|\cdots, n_{\mathbf{p}}, \cdots\rangle = (-1)^{\sum_{j<p} n_j}\,[1-n_{\mathbf{p}}]|\cdots, n_{\mathbf{p}}+1, \cdots\rangle. \tag{A.2a}$$

The factor $(-1)^{\sum_{j<p} n_j}$ is needed to fix the sign of the many-fermion wave function.

Actually, fermions have a nonzero spin. Each state is characterized by a wave vector $\mathbf{p}$ and a spin index σ. The creation and destruction operators thus become $c_{\mathbf{p}\sigma}^{+}$ and $c_{\mathbf{p}\sigma}$. They satisfy anticommutation relations similar to Eq. (A.6):

$$\begin{aligned}[c_{\mathbf{p}\sigma}, c_{\mathbf{p}'\sigma'}]_{+} &= [c_{\mathbf{p}\sigma}^{+}, c_{\mathbf{p}'\sigma'}^{+}]_{+} = 0,\\ [c_{\mathbf{p}\sigma}^{+}, c_{\mathbf{p}'\sigma'}]_{+} &= \delta_{\mathbf{p}\mathbf{p}'}\delta_{\sigma\sigma'}.\end{aligned} \tag{A.6a}$$

The normalized wave function for a plane wave of momentum $\mathbf{p}$ is

$$\frac{1}{\sqrt{\Omega}}e^{i\mathbf{p}\cdot\mathbf{r}}.$$

The probability amplitude for the destruction of a boson of momentum $\mathbf{p}$ taking place at the point $\mathbf{r}$ is thus represented by the operator

$$\frac{1}{\sqrt{\Omega}}e^{i\mathbf{p}\cdot\mathbf{r}}a_{\mathbf{p}}.$$

(with a corresponding expression for the case of a fermion). This leads us to introduce the destruction operator at the point $\mathbf{r}$, $\psi(\mathbf{r})$, defined by

$$\begin{aligned}\psi(\mathbf{r}) &= \frac{1}{\sqrt{\Omega}}\sum_{\mathbf{p}} a_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{r}},\\ a_{\mathbf{p}} &= \frac{1}{\sqrt{\Omega}}\int d^3\mathbf{r}\,\psi(\mathbf{r})e^{-i\mathbf{p}\cdot\mathbf{r}}.\end{aligned} \tag{A.7}$$

The operator $\psi(\mathbf{r})$ describes the destruction at $\mathbf{r}$ of a particle of any momentum whatever. Similarly, the creation operator $\psi^{+}(\mathbf{r})$ is defined by

$$\psi^{+}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}}\sum_{\mathbf{p}} a_{\mathbf{p}}^{+}e^{-i\mathbf{p}\cdot\mathbf{r}}.$$

The commutation relations which ψ and ψ^{+} satisfy are easily deduced from Eqs. (A.3) and (A.4) or from (A.6). They can be written as follows:

$$\begin{aligned}&\text{Bosons,} && [\psi(\mathbf{r}), \psi(\mathbf{r}')] = [\psi^{+}(\mathbf{r}), \psi^{+}(\mathbf{r}')] = 0,\\ & && [\psi(\mathbf{r}), \psi^{+}(\mathbf{r}')] = \delta(\mathbf{r}-\mathbf{r}');\\ &\text{Fermions:} && [\psi_{\sigma}(\mathbf{r}), \psi_{\sigma'}(\mathbf{r}')]_{+} = [\psi_{\sigma}^{+}(\mathbf{r}), \psi_{\sigma'}^{+}(\mathbf{r}')]_{+} = 0,\\ & && [\psi_{\sigma}(\mathbf{r}), \psi_{\sigma'}^{+}(\mathbf{r}')]_{+} = \delta(\mathbf{r}-\mathbf{r}')\delta_{\sigma,\sigma'}.\end{aligned} \tag{A.8}$$

In the case of fermions we have explicitly introduced spin coordi-

nates. It is important not to confuse the *operators* ψ, ψ^+ with the state *vectors* $|\varphi\rangle$.

For the sake of simplicity, we shall consider only Fermi gases. Just as the operator $N_{\mathbf{p}} = c_{\mathbf{p}}^+ c_{\mathbf{p}}$ represents the number of particles in the state $\mathbf{p}$, the operator

$$\rho(\mathbf{r}, \sigma) = \psi_\sigma^+(\mathbf{r})\psi_\sigma(\mathbf{r}) \tag{A.9}$$

represents the density of particles at the point $\mathbf{r}$. The total number of particles in the system is given by

$$N = \sum_\sigma \int d^3\mathbf{r}\psi_\sigma^+(\mathbf{r})\psi_\sigma(\mathbf{r}) = \sum_{\mathbf{p}\sigma} c_{\mathbf{p}\sigma}^+ c_{\mathbf{p}\sigma}. \tag{A.10}$$

These results are extended without difficulty to the study of other physical properties of the system. For example, the kinetic energy

$$T = \sum_i \frac{p_i^2}{2m}$$

can be written in the form

$$T = \sum_{\mathbf{p},\sigma} \frac{p^2}{2m} N_{\mathbf{p}\sigma} = \sum_{\mathbf{p},\sigma} \frac{p^2}{2m} c_{\mathbf{p}\sigma}^+ c_{\mathbf{p}\sigma}. \tag{A.11}$$

By a Fourier transformation, we can go back to ψ, ψ^+. Equation (A.11) then becomes

$$T = -\frac{\hbar^2}{2m} \sum_\sigma \int \psi_\sigma^+(\mathbf{r})\nabla^2\psi_\sigma(\mathbf{r})\, d^3\mathbf{r} = \frac{\hbar^2}{2m} \sum_\sigma \int \nabla\psi_\sigma^+(\mathbf{r}) \cdot \nabla\psi_\sigma(\mathbf{r})\, d^3\mathbf{r}. \tag{A.12}$$

Let us turn now to the study of the potential energy V, which we assume results from a binary interaction, depending only on position,

$$V = \frac{1}{2} \sum_{i \neq j} V(\mathbf{r}_i - \mathbf{r}_j).$$

On using the expansion for a gas of point particles,

$$\rho(\mathbf{r}) = \sum_i \delta(\mathbf{r} - \mathbf{r}_i),$$

we can rewrite the interaction in the form

$$\begin{aligned} V &= \tfrac{1}{2} \int\int d^3\mathbf{r}\, d^3\mathbf{r}'\, V(\mathbf{r} - \mathbf{r}')[\rho(\mathbf{r})\rho(\mathbf{r}') - \rho(\mathbf{r})\delta(\mathbf{r} - \mathbf{r}')], \\ \rho(\mathbf{r}) &= \sum_\sigma \rho(\mathbf{r}, \sigma), \end{aligned} \tag{A.13}$$

where the second term eliminates the interaction of a charge with itself. By using (A.9) and the commutation relations, we can transform (A.13) into

$$V = \frac{1}{2} \sum_{\sigma,\sigma'} \int\int d^3\mathbf{r}\, d^3\mathbf{r}'\, V(\mathbf{r} - \mathbf{r}')\psi_\sigma^+(\mathbf{r})\psi_{\sigma'}^+(\mathbf{r}')\psi_{\sigma'}(\mathbf{r}')\psi_\sigma(\mathbf{r}). \tag{A.14}$$

In general we shall prefer an expression using the operators $c_{\mathbf{p}}$ rather than the operators $\psi(\mathbf{r})$. Let us introduce the Fourier transform of the binary potential,

$$V_{\mathbf{q}} = \int d^3\mathbf{r}\, e^{-i\mathbf{q}\cdot\mathbf{r}} V(\mathbf{r}). \tag{A.15}$$

Equation (A.14) can then be written as

$$V = \frac{1}{2\Omega} \sum_{\mathbf{pkq},\sigma,\sigma'} V_{\mathbf{q}} c_{\mathbf{p}+\mathbf{q},\sigma}^+ c_{\mathbf{k}-\mathbf{q},\sigma'}^+ c_{\mathbf{k},\sigma'} c_{\mathbf{p},\sigma}. \tag{A.16}$$

From Eq. (A.16) and the commutation relations, (A.6a), it follows that the matrix element for scattering a pair of fermions $(\mathbf{p}, \sigma; \mathbf{k}, \sigma')$ to new states $(\mathbf{p} + \mathbf{q}, \sigma; \mathbf{k} - \mathbf{q}, \sigma')$ is given by

$$\begin{aligned} \langle \cdots; \mathbf{p} + \mathbf{q}, \sigma; \mathbf{k} - \mathbf{q}, \sigma'; \cdots |V| \cdots; \mathbf{p}, \sigma; \mathbf{k}; \sigma'; \cdots \rangle \\ = V_{\mathbf{q}} - V_{\mathbf{k}-\mathbf{q}-\mathbf{p}}\delta_{\sigma\sigma'}, \end{aligned} \tag{A.17}$$

provided the states $\mathbf{p}, \mathbf{k}$ are occupied, and those $\mathbf{p} + \mathbf{q}$, $\mathbf{k} - \mathbf{q}$ are empty. The first term is that for direct scattering, the second corresponds to an exchange scattering of the pair, and occurs only if the particles have the same spin. The analogous expression for spin zero bosons is

$$V_{\mathbf{q}} + V_{\mathbf{k}-\mathbf{p}-\mathbf{q}}; \tag{A.18}$$

the difference is again a matter of statistics, that is, of the symmetry of the many-particle wave function.

It is easy to calculate in the same way the expression for any physical quantity in this representation (current, etc.). For example, let us consider the Fourier component of the density,

$$\rho_{\mathbf{q}} = \int d^3\mathbf{r}\, \rho(\mathbf{r}) e^{-i\mathbf{q}\cdot\mathbf{r}}. \tag{A.19}$$

Because of Eqs. (A.7) and (A.9), this can be written as

$$\rho_{\mathbf{q}} = \sum_{\mathbf{p},\sigma} c_{\mathbf{p}\sigma}^+ c_{\mathbf{p}+\mathbf{q},\sigma}. \tag{A.20}$$

In the same way, the current density, $\mathbf{J}_\mathbf{q}$, becomes

$$\mathbf{J}_\mathbf{q} = \sum_{\mathbf{p},\sigma} \frac{\mathbf{p}}{m} c^+_{\mathbf{p}\sigma} c_{\mathbf{p}+\mathbf{q},\sigma}. \tag{A.21}$$

In conclusion, let us remind the reader of the rules for quantizing lattice waves (phonons). The basic Hamiltonian for the lattice waves is

$$H = \sum_\mathbf{q} \frac{P_\mathbf{q}^+ P_\mathbf{q}}{2} + \omega_\mathbf{q}^2 \frac{Q_\mathbf{q}^+ Q_\mathbf{q}}{2}, \tag{A.22}$$

where $P_\mathbf{q}$ and $Q_\mathbf{q}$ are the momentum and coordinate, respectively, of the longitudinal normal modes of the lattice field. They satisfy the harmonic oscillator commutation rules,

$$\begin{aligned} [P_\mathbf{q}, P_{\mathbf{q}'}] &= [Q_\mathbf{q}, Q_{\mathbf{q}'}] = 0, \\ [P_\mathbf{q}, Q_{\mathbf{q}'}] &= -i\delta_{\mathbf{q}\mathbf{q}'}. \end{aligned} \tag{A.23}$$

The transformation to a *phonon* representation, in which the operators which act to create or destroy a phonon, $a_\mathbf{q}^+$ and $a_\mathbf{q}$, are introduced explicitly, is given by

$$\begin{aligned} P_\mathbf{q} &= i\left(\frac{\omega_\mathbf{q}}{2}\right)^{1/2} (a_\mathbf{q}^* - a_{-\mathbf{q}}), \\ Q_\mathbf{q} &= \frac{1}{(2\omega_\mathbf{q})^{1/2}} (a_\mathbf{q} + a^+_{-\mathbf{q}}). \end{aligned} \tag{A.24}$$

The reader may easily verify that the operators $a_\mathbf{q}$ and $a_\mathbf{q}^+$ satisfy the boson commutation relations, (A.1), and that the Hamiltonian (A.22) takes the form

$$H = \sum_\mathbf{q} \omega_\mathbf{q}(a_\mathbf{q}^+ a_\mathbf{q} + \tfrac{1}{2}). \tag{A.25}$$

The interaction between electrons and lattice vibrations thus takes the form (in the second quantized representation)

$$H_{\text{int}} = \sum_\mathbf{q} v_\mathbf{q}^i \rho_\mathbf{q}^+ Q_\mathbf{q} = \sum_{\mathbf{q}\mathbf{p}\sigma} \frac{v_\mathbf{q}^i}{\sqrt{2\omega_\mathbf{q}}} c^+_{\mathbf{p}+\mathbf{q},\sigma} c_{\mathbf{p}\sigma} (a_\mathbf{q} + a^+_{-\mathbf{q}}). \tag{A.26}$$

Let us emphasize that we have discovered nothing new in this appendix: we have just chosen a particularly convenient representation for treating the problems of interest to us.

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